1. State Representation

1. No, has not all transitions are identified and the time of day is a needed variable. To make it Markovian there would need to be all possible combinations of for example, hour in the day, weather and battery state.

2. We can represent this situation by adding an absorbing state S_5 (Broken):

$$P = \begin{array}{c} \\ \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{array}{ccccc} S_1 & S_2 & S_3 & S_4 & S_5 \end{array} \\ \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 & 0.0 \\ 0.2 & 0.3 & 0.2 & 0.1 & 0.0 \\ 0.1 & 0.4 & 0.4 & 0.1 & 0.0 \\ 0.0 & 0.2 & 0.1 & 0.2 & 0.5 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix} \end{array} \begin{array}{c} \\ S_1 \\ S_2 \\ S_3 \\ S_4 \\ S_5 \end{array}$$

3. The new state space could look something like: $\{S_1, S_2, S_3, S_4, S_{1_s}, S_{2_s}, S_{3_s}, S_{4_s}\}$ where the suffix "s" denotes a permanent "slow charging" condition.

The transition probabilities changes would be:

- From S_4 you would have a possibility of transitioning to the slow states
- The slow states are absorbing in the sense that if we transition to any S_{1_s} to S_{4_s} state we would not be able to transition back into a state from S_1 to S_4
- Transitions between $S_{1_s} - S_{4_s}$ are the same as transitions between $S_1 - S_4$

2.-

2.1- $V_2 = V_0 \cdot P^2 \rightarrow V_2 = V_0 \cdot P^2$ $V_0 = [1,0,0,0]$ (first row) $P^2 = P \cdot P$ $V_2 = [0.26 \;\; 0.44 \;\; 0.24 \;\; 0.06]$

$$P = \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix}$$

$$P \cdot P = \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix} \cdot \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.4 & 0.1 \\ 0.0 & 0.5 & 0.1 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.26 & 0.44 & 0.24 & 0.06 \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix}$$

2.2- $V_{10} = V_0 \cdot P^{10}$ $V_0 = [1,0,0,0]$

By using numpy, we create a matrix P and V_0, as well as computing P^{10} with a linear algebra function called matrix_power (P, 10).
Then, V_{10} is computed with $V_0 \cdot P^{10}$ (state distribution calculation at $t=10$), resulting in [0.191711 0.457264 0.235583 0.115442].
From that, we conclude that the probability of the system being in Blackout state (S4) at $t=10$ is $\approx 11.54\%$

2.3- Stationary distribution $\rightarrow \pi \cdot P = \pi$ and $\sum_i \pi_i = 1$ First, we turn $\pi \cdot P = \pi$ into a linear system:

$$\pi \cdot (P - I) = 0$$

Using numpy, we compute $(P-I)^T$. We transpose it so that we can use an algebra function that resolves the 4x4 system with gaussian elimination (solve), that requires π to be placed as follows:

$$(P-I) \cdot \pi = 0$$

And as we know from linear algebra, $(A \cdot B)^T = B^T \cdot A^T$, meaning $(\pi \cdot (P-I))^T = 0 \Leftrightarrow (P-I)^T \cdot \pi^T = 0$.
Following that, we replace the last row with the normalization constraint, ensuring 4 independent equations and that row 4 satisfies $\sum_i \pi_i = 1$.
Lastly, we apply solve to our cases which, after computation, gives us the following:

	S_1	S_2	S_3	S_4	S_5	Conclusions
Original	0.191686	0.457275	0.235566	0.115473	—	Yes (unique)
1.2	0	0	0	0	1	Yes (not unique)
~~1.3~~	~~0.160565~~	~~0.38322~~	~~0.23444 3561 5838~~	~~0.0106 13~~		~~Yes (unique)~~

	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	Conclusions
1.3	0	0	0	0	0.054745	0.383212	0.437007	0.135036	Yes (unique)